



Barker
College

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Student Number

2021

TRIAL HIGHER SCHOOL
CERTIFICATE EXAMINATION

Mathematics Extension 2

Section I - Multiple Choice

Write your student number at the top of this page.

Select the alternative A, B C or D that best answers the question. Fill in the response oval completely.

Sample $2 + 4 =$ A 2 B 6 C 8 D 9
A ☐ B ☒ C ☐ D ☐

If you think you have made a mistake, put a cross through the incorrect answer and fill in the new answer.

A ☒ B ☒ C ☐ D ☐

If you change your mind and have crossed out what you consider to be the correct answer, then indicate this by writing the word *correct* and drawing an arrow as follows.

A ☒ B ☒ C ☐ D ☐
correct

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Start
Here →

1. A ☐ B ☐ C ☐ D ☐

2. A ☐ B ☐ C ☐ D ☐

3. A ☐ B ☐ C ☐ D ☐

4. A ☐ B ☐ C ☐ D ☐

5. A ☐ B ☐ C ☐ D ☐

6. A ☐ B ☐ C ☐ D ☐

7. A ☐ B ☐ C ☐ D ☐

8. A ☐ B ☐ C ☐ D ☐

9. A ☐ B ☐ C ☐ D ☐

10. A ☐ B ☐ C ☐ D ☐

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Student Number

2021

TRIAL HIGHER SCHOOL
CERTIFICATE EXAMINATION

Mathematics Extension 2

Staff Involved:

BHC ARM*

40 copies

8:30 AM
FRIDAY 20 AUGUST

TOPICS COVERED

- Complex Numbers I & II
- Proof
- Integration
- Vectors
- Mechanics

General

Instructions:

- Reading time – 10 minutes
- Working time – 3 hours
- Write using black pen
- Calculators approved by NESA may be used
- A reference sheet is provided separately
- For questions in Section II, show relevant mathematical reasoning and/or calculations
- Marks may not be awarded for careless or poorly arranged working
- Diagrams are not to scale unless specifically stated

Total marks:
100

Section I – 10 marks (pages 4-8)

- Attempt Questions 1-10
- Allow about 15 minutes for this section

Section II – 90 marks (pages 9-14)

- Attempt Questions 11-16
- Allow about 2 hour and 45 minutes for this section

Section I

10 marks

Attempt Questions 1–10

Allow about 15 minutes for this section

Use the multiple-choice answer sheet for Questions 1–10.

- 1 What is the negation of the following statement?

If today is Friday, then I have a maths exam.

- A. If I have a maths exam, then today is Friday.
- B. Today is not Friday and I do not have a maths exam.
- C. If I do not have a maths exam, then today is not Friday.
- D. Today is Friday and I do not have a maths exam.

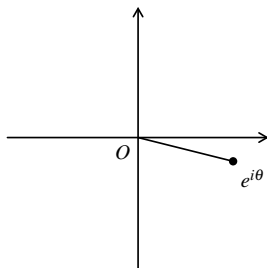
- 2 Given the vectors $\underline{a} = 3\underline{i} - 4\underline{j} + 12\underline{k}$ and $\underline{b} = 2\underline{i} + 2\underline{j} - \underline{k}$, what is the projection of $\underline{a}$ onto $\underline{b}$?

- A. $-\frac{14}{3}(2\underline{i} + 2\underline{j} - \underline{k})$
- B. $-\frac{14}{9}(2\underline{i} + 2\underline{j} - \underline{k})$
- C. $-\frac{14}{13}(3\underline{i} - 4\underline{j} + 12\underline{k})$
- D. $-\frac{14}{169}(3\underline{i} - 4\underline{j} + 12\underline{k})$

- 3 A particle has a displacement of $3\underline{i} + \underline{j}$ metres at a particular time. The particle moves with constant velocity and two seconds later its displacement is $-\underline{i} + 5\underline{j}$ metres. What is the velocity of the particle in m.s^{-1} ?

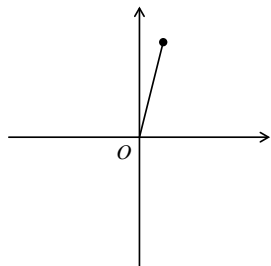
- A. $2\underline{i} + 6\underline{j}$
- B. $-4\underline{i} + 4\underline{j}$
- C. $4\underline{i} - 4\underline{j}$
- D. $-2\underline{i} + 2\underline{j}$

- 4 The Argand diagram shows the complex number $e^{i\theta}$.

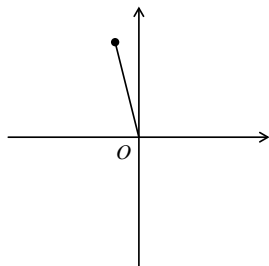


Which of the following diagrams best shows the complex number $ie^{-i\theta}$?

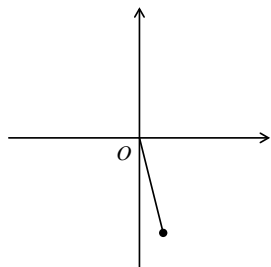
A.



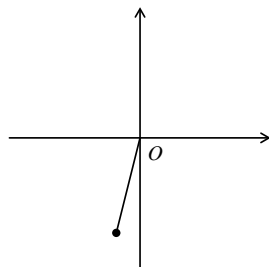
B.



C.



D.



- 5 Given that a and b are integers, which one of the following statements has a converse that is always true?

- A. If a and b are even, then $a + b$ is even
- B. If a is divisible by 8, then a is even
- C. If a and b are odd, then $a \times b$ is odd
- D. If a and b are odd, then $a - b$ is even

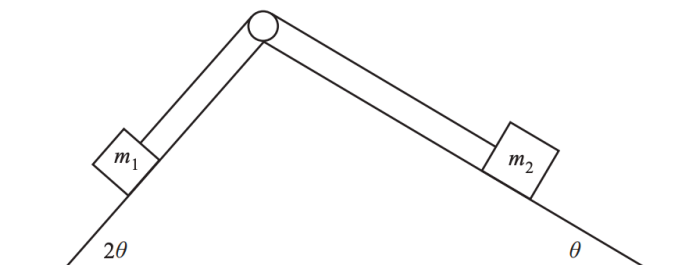
- 6 Which complex number is a cube root of $-i$?

- A. $\frac{\sqrt{3}}{2} - \frac{1}{2}i$
- B. $-\frac{\sqrt{3}}{2} + \frac{1}{2}i$
- C. $\frac{1}{2} - \frac{\sqrt{3}}{2}i$
- D. $-\frac{1}{2} + \frac{\sqrt{3}}{2}i$

- 7 How many complex numbers satisfy $|z - 2 - i| = 3$ and $|z - 6 + 2i| = 2$ simultaneously?

- A. 0
- B. 1
- C. 2
- D. Infinitely many

- 8 Two particles with mass m_1 kg and m_2 kg are connected by a taut light string that passes over a smooth pulley. The particles sit on smooth inclined planes as shown in the diagram below.



If the system is in equilibrium, what is the value of $\frac{m_1}{m_2}$?

- A. $\frac{\sec \theta}{2}$
- B. $2 \sec \theta$
- C. $2 \cos \theta$
- D. $\frac{1}{2}$
- 9 The integral $\int \frac{dx}{2 \sin x + 3 \cos x}$ is equivalent to which one of the following expressions?

- A. $-2 \int \frac{dt}{3t^2 - 4t - 3}$
- B. $-\int \frac{dt}{3t^2 - 4t - 3}$
- C. $-2 \int \frac{dt}{3t^2 - 2t - 3}$
- D. $-\int \frac{dt}{3t^2 - 2t - 3}$

- 10 Given that $\int_a^b f(x) dx > \int_a^b g(x) dx > 0$, which one of the following must be true?

- A. $\forall x \in [a, b] (f(x) > g(x))$
- B. $\forall x \in [a, b] (g(x) > 0)$
- C. $\exists x \in [a, b] (f(x) > g(x))$
- D. $\exists x \in [a, b] (g(x) > f(x))$

Section II

90 marks

Attempt Questions 11–16

Allow about 2 hours and 45 minutes for this section

Answer each question in a SEPARATE writing booklet. Extra writing booklets are available.

For questions in Section II, your responses should include relevant mathematical reasoning and/or calculations.

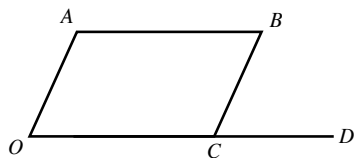
Question 11 (15 marks) Use a SEPARATE Writing Booklet

- (a) Consider the complex numbers $w = 1 - 3i$ and $z = -2 + 5i$.
- (i) Evaluate $z\bar{z}$. 1
- (ii) Express $\frac{w}{z}$ in the form $x + iy$ where x and y are real numbers. 1
- (b) Evaluate $\int_1^e \frac{1}{x} \cos(\ln x) dx$. 2
- Give your answer in simplest exact form.
- (c) Find a vector perpendicular to both $3\hat{i} - 2\hat{j} + \hat{k}$ and $4\hat{i} + 2\hat{j} - 5\hat{k}$. 2
- (d) (i) Find the square root of $5 - 12i$. 2
- (ii) Hence, solve $z^2 + 3z + 1 + 3i = 0$. 2
- (e) Find $\int \frac{dx}{\sqrt{3 - 2x - x^2}}$. 2
- (f) $OABC$ is a parallelogram. Let $\vec{OA} = \vec{a}$ and $\vec{OC} = \vec{c}$. 3

X is the midpoint of AC .

D lies on OC produced such that $\frac{OC}{CD} = k$.

If $\vec{XD} = 3\vec{c} - \frac{1}{2}\vec{a}$, find the value of k .



Question 12 (15 marks) Use a SEPARATE Writing Booklet

- (a) Find the equation, in vector form, of the tangent to the circle $|y| = 5$ at the point $(-3, 4)$. 2
- (b) Using the substitution $x = 4\sin^2 \theta$, evaluate $\int_0^2 \sqrt{\frac{x}{4-x}} dx$. 3
- Give your answer in simplest exact form.
- (c) For positive real numbers a and b , prove that $\frac{1}{a} + \frac{1}{b} \geq \frac{4}{a+b}$. 2
- (d) Consider the following statement, P , for integers x and y :
- P : If $x^2 + y^2$ is divisible by 4 then x and y are not both odd.
- (i) State the contrapositive of P . 1
- (ii) Hence, prove P by using the contrapositive. 2
- (iii) Without providing a proof, state the conditions on x and y for $x^2 + y^2$ to be divisible by 4. 1
- (e) Use a suitable substitution to evaluate $\int_1^9 \frac{dx}{x(2\sqrt{x}-1)}$. 4
- Give your answer in simplest exact form.

Question 13 (14 marks) Use a SEPARATE Writing Booklet

- (a) On an Argand diagram, the point P representing the complex number z moves such that $|z - (1 + i)| = 1$.
- (i) Sketch the locus of P . 1
- (ii) Find the greatest value of $|z|$. 1
- (iii) Shade the region where $|z - (1 + i)| \leq 1$ and $0 \leq \text{Arg}(z - 1) \leq \frac{\pi}{4}$. 2
- (iv) Find the area of the region in (iii). 1
- (b) A ball of mass m kg is thrown vertically downwards off a very high cliff with speed 20 m.s^{-1} . It experiences acceleration due to gravity of 10 m.s^{-2} and a force due to air resistance of $\frac{1}{100}mv^2$ where v is the velocity of the ball.
- (i) Find the terminal velocity that the ball would reach in m.s^{-1} . 1
- (ii) Find the distance through which the ball falls to reach a speed of 30 m.s^{-1} . 3
Give your answer in metres in simplest exact form.
- (c) A particle moves in a straight line. Its displacement, x metres, after t seconds is given by $x = 2 \cos\left(2t - \frac{2\pi}{3}\right) + 3$.
- (i) Prove the particle is moving in simple harmonic motion about $x = 3$ by showing that $\ddot{x} = -4(x - 3)$. 2
- (ii) What is the period of the motion? 1
- (iii) Find all times within the first π seconds when the particle is moving at 2 metres per second in either direction. 2

Question 14 (15 marks) Use a SEPARATE Writing Booklet

- (a) A particle of mass m kg is projected vertically upwards with a velocity of $\sqrt{\frac{g}{k}} \text{ m.s}^{-1}$ 4
where g is the magnitude of the acceleration due to gravity. It experiences air resistance of magnitude mkv^2 . Show that the time taken to reach its maximum height is $\frac{\pi}{4\sqrt{gk}}$.
- (b) The point A has a position vector $\begin{pmatrix} -2 \\ 4 \\ 7 \end{pmatrix}$ and the point B has position vector $\begin{pmatrix} -1 \\ 3 \\ 8 \end{pmatrix}$.
- (i) Find a vector equation for ℓ_1 , the line passing through A and B . 2
- (ii) The point P has position vector $\begin{pmatrix} 0 \\ 2 \\ 3 \end{pmatrix}$. 2
If $\angle PBA = \theta$, show that $\cos \theta = \frac{1}{3}$.
- (iii) The line ℓ_2 passes through the point P and is parallel to the line ℓ_1 . 1
Find a vector equation for ℓ_2 .
- (iv) The points C and D both lie on ℓ_2 . Given that $AB = PC = DP$ and that the x -coordinate of C is positive, find the coordinates of C and D . 3
- (v) Find the exact area of trapezium $ABCD$, giving your answer as a simplified surd. 3

Question 15 (17 marks) Use a SEPARATE Writing Booklet

- (a) A particle is moving such that $\ddot{x} = 18x^3 + 27x^2 + 9x$.
Initially, $x = -2$ and the velocity, $v = -6$.
- (i) Show that $v^2 = 9x^2(1+x)^2$. 2
- (ii) Hence, or otherwise, show that $\int \frac{1}{x(1+x)} dx = -3t$. 2
- (iii) Hence, find x as a function of t . 4
- (b) Find $\int (1+2x^2)e^{x^2} dx$. 3
- (c) (i) Use de Moivre's Theorem to show that 3
- $$\tan 7\theta = \frac{7t - 35t^3 + 21t^5 - t^7}{1 - 21t^2 + 35t^4 - 7t^6} \quad \text{where } t = \tan \theta$$
- (ii) Hence, show that the roots of the equation $x^3 - 21x^2 + 35x - 7 = 0$ are 3
- $$\tan^2\left(\frac{\pi}{7}\right), \tan^2\left(\frac{2\pi}{7}\right) \text{ and } \tan^2\left(\frac{3\pi}{7}\right)$$

Question 16 (14 marks) Use a SEPARATE Writing Booklet

- (a) (i) For a function $g(x)$, continuous on $[a, b]$, prove that 2
- $$\int_a^b g(x) dx = \int_a^b g(a+b-x) dx$$
- (ii) Hence, or otherwise, for an **even** function $f(x)$, continuous on $[-a, a]$, show that 3
- $$\int_{-a}^a \frac{f(x)}{1+e^x} dx = \int_0^a f(x) dx$$
- (b) For integers $n \geq 0$, let $I_n(x) = \int_0^x t^n e^{-t} dt$.
- (i) Prove by induction that 4
- $$I_n(x) = n! \left[1 - e^{-x} \left(1 + x + \frac{x^2}{2!} + \frac{x^3}{3!} + \cdots + \frac{x^n}{n!} \right) \right]$$
- (ii) Show that 2
- $$0 \leq \int_0^1 t^n e^{-t} dt \leq \frac{1}{n+1}$$
- (iii) Hence, show that 1
- $$0 \leq 1 - e^{-1} \left(1 + \frac{1}{1!} + \frac{1}{2!} + \cdots + \frac{1}{n!} \right) \leq \frac{1}{(n+1)!}$$
- (iv) Hence, find the limiting value of $1 + \frac{1}{1!} + \frac{1}{2!} + \cdots + \frac{1}{n!}$ as $n \rightarrow \infty$. 2
- Show working to justify your answer.

End of paper

Year 12 Extension 2 Trial Exam

Q1, "If today is Friday, then I have a maths exam" is of the form

$$F \Rightarrow E$$

which is logically equivalent to

$$\sim F \vee E.$$

The negation of $F \Rightarrow E$ is

$$\sim(\sim F \vee E)$$

which is equivalent to

$$F \wedge \sim E,$$

namely,

"It is Friday and I don't have a maths exam."

(d)

Q2, Now $\text{proj}_{\vec{b}} \vec{a}$

$$= \frac{\vec{a} \cdot \vec{b}}{|\vec{b}|^2} \times \vec{b}$$

$$= \frac{6-8-12}{(\sqrt{9})^2} \times (2\vec{i} + 2\vec{j} - \vec{k})$$

$$= -\frac{14}{9} (2\vec{i} + 2\vec{j} - \vec{k})$$

(b)

Q3, The displacement in 2 sec is

$$(\vec{i} + 5\vec{j}) - (3\vec{i} + \vec{j}) = -4\vec{i} + 4\vec{j}$$

Hence the velocity in m/s is

$$\frac{-4\vec{i} + 4\vec{j}}{2} = -2\vec{i} + 2\vec{j}$$

(d)

Q4, Let $\alpha = -\frac{\pi}{6}$ such that

$$e^{i\alpha} = e^{i(-\pi/6)} \text{ then}$$

$$ie^{i\alpha} = ie^{i(-\pi/6)}$$

$$= i\left(\cos\frac{\pi}{6} + i\sin\frac{\pi}{6}\right)$$

$$= -\frac{1}{2} + \frac{\sqrt{3}}{2}i$$

(b)

Q5, The converse of $p \Rightarrow q$ is

$$q \Rightarrow p.$$

The converse of (c) is

"If $a \times b$ is odd, then

a and b are odd."

This is always true.

$$Q6, \vec{z}^3 = -i$$

$$= \cos\left(-\frac{\pi}{2}\right) + i\sin\left(-\frac{\pi}{2}\right)$$

$$\therefore \vec{z} = \cos\left(-\frac{\pi}{6}\right) + i\sin\left(-\frac{\pi}{6}\right)$$

$$\vec{z} = \frac{\sqrt{3}}{2} - \frac{1}{2}i$$

(a)

$$Q7, |z - (2+i)| = 3$$

is a circle centre $(2,1)$
radius 3 units.

$$|z - (6-2i)| = 2$$

is a circle centre $(6,-2)$
radius 2 units.

Consider the distance between the two centres:

$$(2,1) \quad d = \sqrt{16+9} = 5$$

$$\text{But } 3+2=5$$

$\Rightarrow$ The two circles touch at one point (b)

$$Q8, \int \frac{dx}{2\sin x + 3\cos x} \quad dx$$

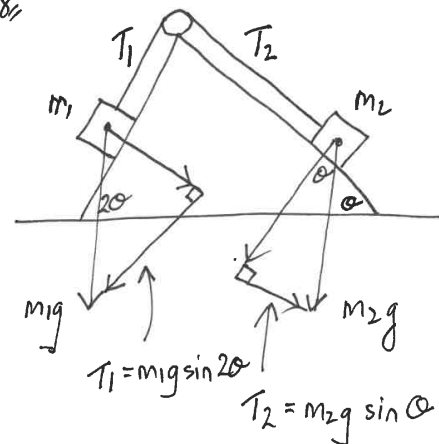
$$\text{Let } t = \tan\left(\frac{x}{2}\right)$$

$$\frac{dt}{dx} = \frac{1}{2} \sec^2\left(\frac{x}{2}\right)$$

$$\frac{dt}{dx} = \frac{1}{2}(1+t^2)$$

$$dx = \frac{2dt}{1+t^2}$$

Q8,



$$T_1 = T_2$$

$$\therefore m_1 g \sin 20 = m_2 g \sin \theta$$

$$\therefore \frac{m_1}{m_2} = \frac{\sin \theta}{\sin 20} = \frac{1}{2 \cos \theta}$$

(a)

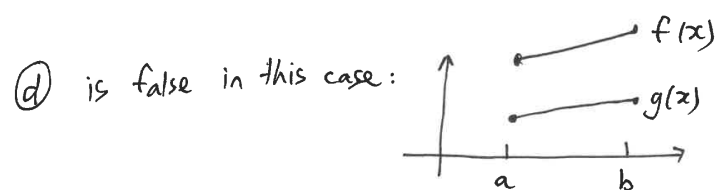
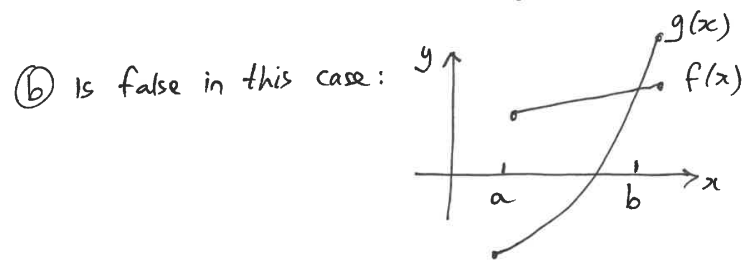
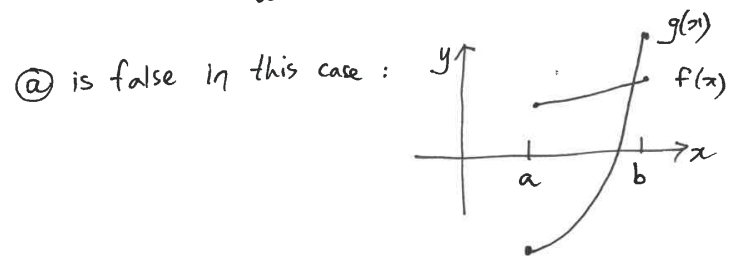
$$I = \int \frac{2dt/1+t^2}{2\left(\frac{2t}{1+t^2}\right) + 3\left(\frac{1-t^2}{1+t^2}\right)}$$

$$= \int \frac{2dt}{4t + 3 - 3t^2}$$

$$= \int \frac{-2dt}{(3t^2 - 4t - 3)}$$

(a)

Q10, Given that $\int_a^b f(x) dx > \int_a^b g(x) dx > 0$



Therefore (c)
because if there was not one single x -value
for which $f(x) > g(x)$ then $f(x)$ lies entirely
underneath $g(x)$, which would make it
impossible that

$$\int_a^b f(x) dx > \int_a^b g(x) dx > 0.$$

Q11,

(a) $w = 1-3i$

$$z = -2+5i$$

(i) $z\bar{z} = (-2+5i)(-2-5i)$
 $= 29$

(ii) $\frac{w}{z} = \frac{1-3i}{-2+5i} \times \frac{-2-5i}{-2-5i}$
 $= \frac{-2-5i+6i-15}{29}$
 $= \frac{-17}{29} + \frac{1}{29}i$

(b) $\int_1^e \frac{1}{x} \cos(\ln x) dx$
 $= \sin(\ln x) \Big|_1^e$
 $= \sin(1)$
 ≈ 0.8415

(c) let the required vector be
expressed as $\begin{pmatrix} a \\ b \\ c \end{pmatrix}$. We need

$$3a - 2b + c = 0 \quad \text{and} \\ 4a + 2b - 5c = 0$$

$$7a - 4c = 0$$

$$a = \frac{4c}{7}$$

If $c=1$, $a = \frac{4}{7}$ and $b = \frac{19}{14}$

(d) (i)

let $(x+iy)^2 = 5-12i$

$$x^2 - y^2 = 5 \quad 2xyi = -12i$$

$$y = \frac{-6}{x}$$

$$x^2 - \left(\frac{-6}{x}\right)^2 = 5$$

$$x^4 - 5x^2 - 36 = 0$$

$$(x^2-9)(x^2+4) = 0$$

$$x = \pm 3$$

when $x=3$, $y=-2$

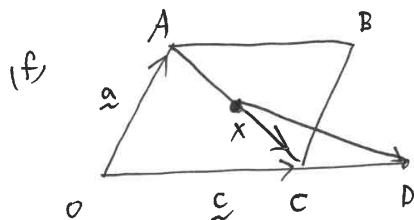
when $x=-3$, $y=2$

Hence $\pm(3-2i)$

(ii) $z^2 + 3z + (1+3i) = 0$
 $z = \frac{-3 \pm \sqrt{5-12i}}{2}$
 $z = \frac{-3 \pm (3-2i)}{2}$
 $z = -i, -3+i$

Hence $\begin{pmatrix} a \\ b \\ c \end{pmatrix} = \begin{pmatrix} \frac{4}{7} \\ \frac{19}{14} \\ 1 \end{pmatrix}$
 $= \begin{pmatrix} 8 \\ 19 \\ 14 \end{pmatrix}$

$$\begin{aligned}
 \text{(e)} \quad & \int \frac{dx}{\sqrt{3-2x-x^2}} \\
 &= \int \frac{dx}{\sqrt{3-(x^2+2x)}} \\
 &= \int \frac{dx}{\sqrt{4-(x+1)^2}} \\
 &= \sin^{-1} \left(\frac{x+1}{2} \right) + C
 \end{aligned}$$

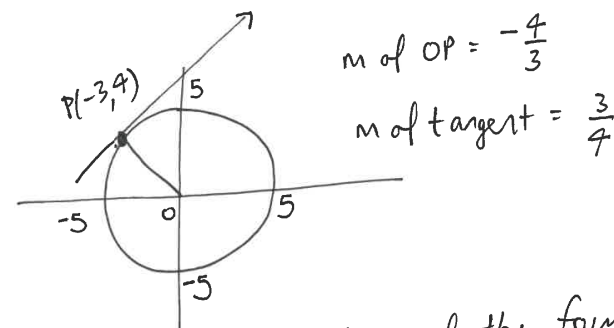


$$\begin{aligned}
 \frac{OC}{CD} &= k \Rightarrow CD = \frac{c}{k} \\
 \vec{XD} &= 3\vec{c} - \frac{1}{2}\vec{a} \\
 \text{Now } \vec{AC} &= \vec{c} - \vec{a} \\
 \vec{OX} &= \vec{OA} + \frac{1}{2}\vec{AC} \\
 &= \vec{a} + \frac{1}{2}(\vec{c} - \vec{a}) \\
 &= \frac{1}{2}(\vec{a} + \vec{c}) \\
 \vec{OD} &= \vec{OX} + \vec{XD} \\
 &= \frac{\vec{a}}{2} + \frac{\vec{c}}{2} + 3\vec{c} - \frac{\vec{a}}{2} \\
 &= \frac{7\vec{c}}{2}
 \end{aligned}$$

But $\vec{OD} = \vec{OC} + \vec{CD}$
 $= \vec{c} + \frac{\vec{c}}{k}$
 $\therefore \frac{7\vec{c}}{2} = \vec{c} + \frac{\vec{c}}{k}$
 $\frac{5\vec{c}}{2} = \frac{\vec{c}}{k}$
 $\therefore k = \frac{2}{5}$

Q12

(a)



The tangent has a vector equation of the form
 $\underline{R} = (-3\hat{i} + 4\hat{j}) + k(4\hat{i} + 3\hat{j})$

$$\text{(b)} \quad \int_0^2 \sqrt{\frac{x}{4-x}} dx$$

$$\begin{aligned}
 \text{Let } x &= 4 \sin^2 \theta \\
 dx &= 8 \sin \theta \cos \theta d\theta
 \end{aligned}$$

$$\begin{aligned}
 \text{When } x=2, \quad \theta &= \frac{\pi}{4} \\
 \text{When } x=0, \quad \theta &= 0
 \end{aligned}$$

$$\begin{aligned}
 I &= \int_0^{\pi/4} \sqrt{\frac{4 \sin^2 \theta}{4 - 4 \sin^2 \theta}} \times 8 \sin \theta \cos \theta d\theta \\
 &= 8 \int_0^{\pi/4} \sin^2 \theta d\theta \\
 &= 4 \int_0^{\pi/4} 1 - \cos 2\theta d\theta \\
 &= 4 \left[\theta - \frac{1}{2} \sin 2\theta \right]_0^{\pi/4} \\
 &= \pi - 2
 \end{aligned}$$

(c) Prove that $\frac{1}{a} + \frac{1}{b} \geq \frac{4}{a+b}$

Consider LHS - RHS

$$= \frac{1}{a} + \frac{1}{b} - \frac{4}{a+b}$$

$$= \frac{b(a+b) + a(a+b) - 4ab}{ab(a+b)}$$

$$= \frac{a^2 - 2ab + b^2}{ab(a+b)}$$

$$= \frac{(a-b)^2}{ab(a+b)}$$

Now the numerator is always ≥ 0 and the denominator is always > 0 .

Hence LHS - RHS ≥ 0 .

Therefore

$$\frac{1}{a} + \frac{1}{b} \geq \frac{4}{a+b}.$$

(d) P: If $x^2 + y^2$ is divisible by 4 then x and y are not both odd.

(i) The contrapositive of P is:

"If x and y are both odd then $x^2 + y^2$ is not divisible by 4."

(ii) Let $x = 2k+1$
 $y = 2j+1$

$$\begin{aligned} \text{then } x^2 + y^2 &= (2k+1)^2 + (2j+1)^2 \\ &= 4k^2 + 4k + 1 + 4j^2 + 4j + 1 \\ &= 4(k^2 + k + j^2 + j) + 2 \end{aligned}$$

which is not divisible by 4.

Hence, if x and y are both odd then $x^2 + y^2$ is not divisible by 4.

Therefore, if $x^2 + y^2$ is divisible by 4, then x and y are not both odd.

(iii) For $x^2 + y^2$ to be divisible by 4, the condition on x and y is that they are both even.

$$(e) \int_1^9 \frac{dx}{x(2\sqrt{x}-1)}$$

$$\text{let } x = u^2 \\ dx = 2u du$$

$$\text{when } x=9, u=3 \\ \text{when } x=1, u=1$$

$$I = \int_1^3 \frac{2u du}{u^2(2u-1)} \\ = \int_1^3 \frac{2 du}{u(2u-1)}$$

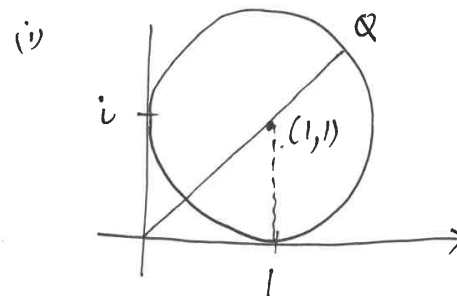
$$\text{let } \frac{2}{u(2u-1)} = \frac{A}{u} + \frac{B}{2u-1}$$

$$2 = A(2u-1) + Bu$$

$$A = -2, B = 4$$

$$I = \int_1^3 \left(\frac{-2}{u} + \frac{4}{2u-1} \right) du \\ = \left[-2 \ln u + 2 \ln(2u-1) \right]_1^3 \\ = -2 \ln 3 + 2 \ln 5 - 0 \\ = \ln \left(\frac{25}{9} \right)$$

$$Q13, (a) |z - (1+i)| = 1$$

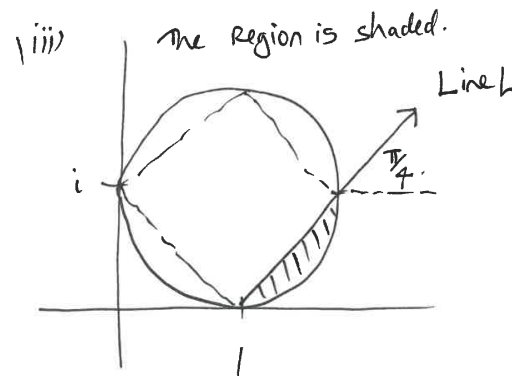


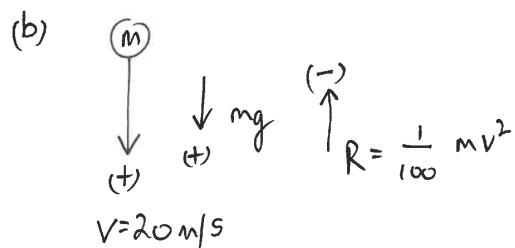
(ii) The greatest value of $|z|$ occurs when P is at Q.

The greatest value of $|z|$ is $1 + \sqrt{2}$

(iv) Line L has equation $y = x - 1$. L intersects the circle at (2,1).

$$\text{Shaded area} \\ = \frac{1}{4} [\pi R^2 - (\sqrt{2})^2] \\ = \frac{1}{4} (\pi - 2)$$





$$(i) \quad m\ddot{x} = 10m - \frac{1}{100} m v^2$$

$$\ddot{x} = 10 - \frac{1}{100} v^2$$

Put $\ddot{x} = 0$,

$$0 = 10 - \frac{1}{100} v^2$$

$$v^2 = 1000$$

$$v = 10\sqrt{10} \text{ m/s.}$$

Terminal velocity is $10\sqrt{10} \text{ m/s.}$

$$(ii) \quad v \frac{dv}{dx} = 10 - \frac{1}{100} v^2$$

$$\frac{dv}{dx} = \frac{10 - \frac{1}{100} v^2}{v}$$

$$\frac{dv}{dx} = \frac{1000 - v^2}{100v}$$

$$\frac{dx}{dv} = \frac{100v}{1000 - v^2}$$

$$x = -50 \int \frac{-2v}{1000 - v^2} dv$$

$$x = -50 \ln(1000 - v^2) + C$$

But when $x = 0$, $v = 20$ so

$$C = 50 \ln 600$$

$$\therefore x = 50 \ln \left(\frac{600}{1000 - v^2} \right)$$

When $v = 30 \text{ m/s,}$

$$x = 50 \ln \left(\frac{600}{100} \right)$$

$$x = 50 \ln(6)$$

$$(c) \quad x = 2 \cos \left(2t - \frac{2\pi}{3} \right) + 3$$

$$(i) \quad \dot{x} = -4 \sin \left(2t - \frac{2\pi}{3} \right)$$

$$\ddot{x} = -8 \cos \left(2t - \frac{2\pi}{3} \right)$$

$$\ddot{x} = -4 \left(2 \cos \left(2t - \frac{2\pi}{3} \right) \right)$$

$$\ddot{x} = -4(x - 3)$$

Hence, the particle is moving in SHM about $x = 3$.

$$(ii) \quad \text{The period is } \frac{2\pi}{2} = \pi \text{ seconds.}$$

(iii) To find all the times in the first π seconds when the particle is travelling at 2 m/s left or right, solve $2 = -4 \sin \left(2t - \frac{2\pi}{3} \right)$ and $-2 = -4 \sin \left(2t - \frac{2\pi}{3} \right)$

$$-\frac{1}{2} = \sin \left(2t - \frac{2\pi}{3} \right)$$

$$\frac{1}{2} = \sin \left(2t - \frac{2\pi}{3} \right)$$

$$\therefore 2t - \frac{2\pi}{3} = \left(\pi + \frac{\pi}{6} \right), \frac{11\pi}{6}, \frac{-\pi}{6} \quad 2t - \frac{2\pi}{3} = \frac{\pi}{6}, \frac{5\pi}{6}$$

$$2t = \frac{11\pi}{6}, \frac{3\pi}{6}$$

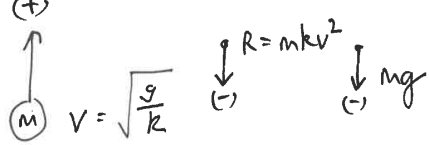
$$2t = \frac{5\pi}{6}, \frac{9\pi}{6}$$

$$t = \frac{11\pi}{12}, \frac{\pi}{4}$$

$$t = \frac{5\pi}{12}, \frac{3\pi}{4}$$

$$\therefore t = \frac{\pi}{4}, \frac{5\pi}{12}, \frac{3\pi}{4}, \frac{11\pi}{12}$$

Q14 (a)



$$m\ddot{x} = -mg - mkv^2$$

$$\ddot{x} = -g - kv^2$$

$$\frac{dv}{dt} = -g - kv^2$$

$$\frac{dt}{dv} = \frac{-1}{g + kv^2}$$

$$t = \frac{-1}{k} \int \frac{dv}{(\frac{g}{k}) + v^2}$$

$$t = \frac{-1}{k} \sqrt{\frac{k}{g}} \tan^{-1} \left(\sqrt{\frac{g}{k}} \frac{v}{1} \right) + C$$

But when $t=0$, $v = \sqrt{g/k}$ so

$$C = \frac{1}{k} \sqrt{\frac{k}{g}} \tan^{-1}(1)$$

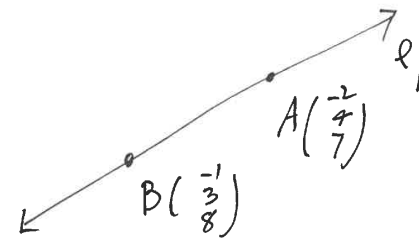
$$C = \frac{1}{\sqrt{gk}} \times \frac{\pi}{4}$$

$$\therefore t = \frac{\pi}{4\sqrt{gk}} - \frac{1}{\sqrt{gk}} \tan^{-1} \left(\sqrt{\frac{g}{k}} \frac{v}{1} \right)$$

when $v=0$,

$$t = \frac{\pi}{4\sqrt{gk}} \text{ as required.}$$

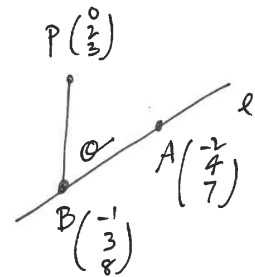
(b) (i)



$$r_1 \text{ is } \vec{OA} + \lambda \vec{AB}$$

$$= \begin{pmatrix} -2 \\ 7 \end{pmatrix} + \lambda \begin{pmatrix} -1 \\ 1 \end{pmatrix}$$

(ii)



$$\text{Now } \vec{BP} = \begin{pmatrix} 1 \\ -1 \\ -5 \end{pmatrix}$$

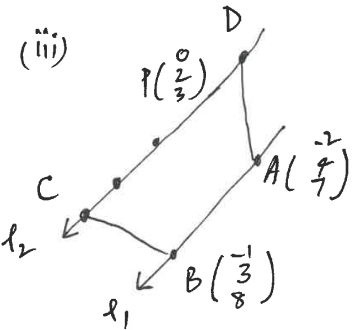
$$\text{and } \vec{BA} = \begin{pmatrix} -1 \\ 1 \\ -1 \end{pmatrix}$$

$$\therefore \vec{BP} \cdot \vec{BA} = -1 - 1 + 5 = 3$$

$$\therefore 3 = \sqrt{27} \times \sqrt{3} \times \cos \theta$$

$$\therefore \cos \theta = \frac{3}{3\sqrt{3} \times \sqrt{3}}$$

$$\cos \theta = \frac{1}{3} \text{ as required.}$$



$$\text{A vector equation for } l_2 \text{ is}$$

$$= \begin{pmatrix} 0 \\ 2 \\ 3 \end{pmatrix} + \lambda \begin{pmatrix} -1 \\ -1 \\ 1 \end{pmatrix}$$

(iv) Now $AB = \sqrt{3}$ so

$$C = \begin{pmatrix} 1 \\ 4 \end{pmatrix} \text{ and}$$

$$D = \begin{pmatrix} -1 \\ 3 \\ 2 \end{pmatrix}$$

$$(v) |CD| = \sqrt{12}$$

$$|AB| = \sqrt{3}$$

$$h = \sqrt{24}$$

$$\text{Area} = 9\sqrt{2}$$

Q15 (a)

$$\dot{x} = 18x^3 + 27x^2 + 9x$$

when $t=0$, $x=-2$ and $v=-6$

$$(i) \frac{d}{dx} \left(\frac{1}{2} v^2 \right) = 18x^3 + 27x^2 + 9x$$

$$\frac{1}{2} v^2 = \frac{9}{2} x^4 + 9x^3 + \frac{9}{2} x^2 + C$$

When $x=-2$, $v=-6$ so

$$18 = 72 - 72 + 18 + C$$

$$C = 0$$

$$\therefore v^2 = 9x^4 + 18x^3 + 9x^2$$

$$v^2 = 9x^2(x^2 + 2x + 1)$$

$$v^2 = 9x^2(x+1)^2 \text{ as required.}$$

$$(ii) \text{ From } v^2 = 9x^2(x+1)^2$$

$$v = \pm 3x(x+1)$$

But when $x=-2$, $v=-6$ so

$$-6 = -3(-2)(-1)$$

$$\therefore v = -3x(x+1)$$

$$\therefore \frac{dx}{dt} = -3x(x+1)$$

$$\frac{dt}{dx} = \frac{1}{-3x(x+1)}$$

$$\int -3 dt = \int \frac{dx}{x(x+1)}$$

$$\therefore \int \frac{dx}{x(x+1)} = -3t$$

as required.

(iii)

$$\frac{dx}{dt} = -3x(x+1)$$

$$t = \int \frac{-1}{3x(x+1)} dx$$

$$t = -\frac{1}{3} \int \frac{1}{x} - \frac{1}{x+1} dx$$

$$t = -\frac{1}{3} [\ln x - \ln(x+1)] + C$$

But when $t=0$, $x=-2$ so

$$0 = -\frac{1}{3} \ln 2 + C$$

$$C = \frac{1}{3} \ln 2$$

$$\therefore t = -\frac{1}{3} \ln \left(\frac{x}{x+1} \right) + \frac{1}{3} \ln 2$$

$$t = \frac{1}{3} \ln \left(\frac{2(x+1)}{x} \right)$$

$$e^{3t} = \frac{2x+2}{x}$$

$$xe^{3t} = 2x+2$$

$$-2 = x(2 - e^{3t})$$

$$\therefore x = \frac{-2}{2 - e^{3t}}$$

$$(b) \int (1+2x^2) e^{x^2} dx$$

$$= \int e^{x^2} + 2x^2 e^{x^2} dx$$

$$= \int e^{x^2} dx + \int x(2xe^{x^2}) dx$$

$$= \int e^{x^2} dx + \left[e^{x^2} \cdot x - \int e^{x^2} \cdot (1) dx \right]$$

$$= \int e^{x^2} dx + x e^{x^2} - \int e^{x^2} dx$$

$$= x e^{x^2}$$

(c) (i)

$$(\cos \theta + i \sin \theta)^7 = \cos 7\theta + i \sin 7\theta \text{ by de Moivre's Theorem}$$

$$\text{Also } (\cos \theta + i \sin \theta)^7 = \cos^7 \theta + 7 \cos^6 \theta (i \sin \theta) + 21 \cos^5 \theta (i \sin \theta)^2 + 35 \cos^4 \theta (i \sin \theta)^3 + 35 \cos^3 \theta (i \sin \theta)^4 + 21 \cos^2 \theta (i \sin \theta)^5 + 7 \cos \theta (i \sin \theta)^6 + (i \sin \theta)^7$$

by Binomial Theorem

$$= \{ \cos^7 \theta - 21 \cos^5 \theta \sin^2 \theta + 35 \cos^3 \theta \sin^4 \theta - 7 \cos \theta \sin^6 \theta \} + i \{ 7 \cos^6 \theta \sin \theta - 35 \cos^4 \theta \sin^3 \theta + 21 \cos^2 \theta \sin^5 \theta - \sin^7 \theta \}$$

$$\tan 7\theta = \frac{\sin 7\theta}{\cos 7\theta}$$

$$= \frac{7\cos^6\theta \sin\theta - 35\cos^4\theta \sin^3\theta + 21\cos^2\theta \sin^5\theta - \sin^7\theta}{\cos^7\theta - 21\cos^5\theta \sin^2\theta + 35\cos^3\theta \sin^4\theta - 7\cos\theta \sin^6\theta}$$

Divide numerator and denominator by $\cos^7\theta$ to get

$$= \frac{7\tan\theta - 35\tan^3\theta + 21\tan^5\theta - \tan^7\theta}{1 - 21\tan^2\theta + 35\tan^4\theta - 7\tan^6\theta}$$

$$= \frac{7t - 35t^3 + 21t^5 - t^7}{1 - 21t^2 + 35t^4 - 7t^6} \quad \text{if } t = \tan\theta.$$

(ii) $\tan 7\theta = 0$ if

$$7t - 35t^3 + 21t^5 - t^7 = 0$$

$$-t(t^6 - 21t^4 + 35t^2 - 7) = 0$$

$$\therefore t = 0 \text{ or } t^6 - 21t^4 + 35t^2 - 7 = 0 \quad \text{--- (1)}$$

Let $t^2 = x$, then (1) becomes

$$x^3 - 21x^2 + 35x - 7 = 0 \quad \text{--- (2)}$$

The solutions to (2) lie among the solutions to

$$\tan 7\theta = 0$$

$$7\theta = 0, \pi, 2\pi, 3\pi, \dots, 6\pi$$

$$\theta = 0, \frac{\pi}{7}, \frac{2\pi}{7}, \frac{3\pi}{7}, \dots, \frac{6\pi}{7}$$

We let $x = t^2$, so $x = \tan^2\theta$

$\therefore$ the solutions to

$$x^3 - 21x^2 + 35x - 7 = 0 \text{ are}$$

$$x = \tan^2\left(\frac{\pi}{7}\right), \tan^2\left(\frac{2\pi}{7}\right), \text{ and } \tan^2\left(\frac{3\pi}{7}\right).$$

Q16, (a) (i)

To prove that $\int_a^b g(x) dx = \int_a^b g(a+b-x) dx$,

consider $I = \int_a^b g(a+b-x) dx$.

$$\text{let } u = a+b-x$$

$$\therefore du = -dx$$

$$\text{when } x = b, \quad u = a$$

$$\text{when } x = a, \quad u = b$$

$$\therefore I = \int_b^a g(u) (-du)$$

$$= \int_a^b g(u) du$$

$$= \int_a^b g(x) dx$$

(u is a dummy variable)

$$\therefore \int_a^b g(x) dx = \int_a^b g(a+b-x) dx$$

as required.

(b) (ii)

To show that $\int_{-a}^a \frac{f(x)}{1+e^x} dx = \int_0^a f(x) dx$.

where $f(x)$ is an even function.

$$\text{Now } \int_{-a}^a \frac{f(x)}{1+e^x} dx$$

$$= \int_{-a}^a \frac{f(a+(-a)-x)}{1+e^{(a-a-x)}} dx$$

$$= \int_{-a}^a \frac{f(-x)}{1+e^{-x}} dx$$

$$= \int_{-a}^a \frac{e^x f(-x)}{1+e^x} dx$$

$$= \int_{-a}^a \frac{(1+e^x)f(-x)}{1+e^x} - \frac{f(-x)}{1+e^x} dx$$

$$= \int_{-a}^a f(-x) - \frac{f(-x)}{1+e^x} dx$$

$$= \int_{-a}^a f(x) - \frac{f(x)}{1+e^x} dx, \text{ since } f(x) = f(-x)$$

$$= \int_{-a}^a f(x) dx - \int_{-a}^a \frac{f(x)}{1+e^x} dx$$

$$\therefore 2 \int_{-a}^a \frac{f(x)}{1+e^x} dx = \int_{-a}^a f(x) dx$$

$$\therefore 2 \int_{-a}^a \frac{f(x)}{1+e^x} dx = 2 \int_0^a f(x) dx$$

$$\therefore \int_{-a}^a \frac{f(x)}{1+e^x} dx = \int_0^a f(x) dx, \text{ as required.}$$

(b) Let $I_n = \int_0^x t^n e^{-t} dt$, $n \geq 0$ (integers)

(i) To prove that $I_n = n! \left[1 - e^{-x} \left(1 + x + \frac{x^2}{2!} + \dots + \frac{x^n}{n!} \right) \right]$

when $n=0$, $I_n = \int_0^x t^0 e^{-t} dt$

$$= \int_0^x e^{-t} dt$$

$$= \left[-e^{-t} \right]_0^x$$

$$= -e^{-x} - (-e^0) = 1 - e^{-x}$$

and also $I_n = 0! \left[1 - e^{-x} (1) \right]$

$$= 1 (1 - e^{-x}) = 1 - e^{-x}$$

Same!

$\therefore$ The statement is true when $n=0$.

Assume that the statement is true when $n=k$, i.e., assume

$$\int_0^x t^k e^{-t} dt = k! \left[1 - e^{-x} \left(1 + x + \dots + \frac{x^k}{k!} \right) \right]$$

and then try to prove that

$$\int_0^x t^{k+1} e^{-t} dt = (k+1)! \left[1 - e^{-x} \left(1 + x + \dots + \frac{x^k}{k!} + \frac{x^{k+1}}{(k+1)!} \right) \right]$$

Now $\int_0^x t^{k+1} e^{-t} dt$

$$= \left[-e^{-t} \cdot t^{k+1} \right]_0^x - \int_0^x -e^{-t} (k+1) t^k dt$$

$$= -x^{k+1} e^{-x} + (k+1) \int_0^x t^k e^{-t} dt$$

$$= -x^{k+1} e^{-x} + (k+1) \left[k! \left(1 - e^{-x} \left(1 + x + \dots + \frac{x^k}{k!} \right) \right) \right]$$

by substituting the assumption.

$$= -x^{k+1} e^{-x} + (k+1)! \left[1 - e^{-x} \left(1 + x + \dots + \frac{x^k}{k!} \right) \right]$$

$$= \frac{-x^{k+1} e^{-x} (k+1)!}{(k+1)!} + (k+1)! \left[1 - e^{-x} \left(1 + x + \dots + \frac{x^k}{k!} \right) \right]$$

$$= (k+1)! \left[1 - e^{-x} \left(1 + x + \dots + \frac{x^k}{k!} + \frac{x^{k+1}}{(k+1)!} \right) \right]$$

= RHS

$\therefore$ The statement is true by mathematical induction.

(ii) Show that

$$0 \leq \int_0^1 t^n e^{-t} dt \leq \frac{1}{n+1}$$

Now $0 \leq e^{-t} \leq 1$ for $0 \leq t \leq 1$

$$0 \leq t^n e^{-t} \leq t^n$$

$$0 \leq \int_0^1 t^n e^{-t} dt \leq \int_0^1 t^n dt$$

$$\text{But } \int_0^1 t^n dt = \left[\frac{t^{n+1}}{n+1} \right]_0^1 = \frac{1}{n+1}$$

$$\therefore 0 \leq \int_0^1 t^n e^{-t} dt \leq \frac{1}{n+1} \text{ as required.}$$

(iii) sub $x=1$ into the result in (i)
and then use (ii) to get

$$0 \leq n! \left[1 - e^{-1} \left(1 + \frac{1}{1!} + \frac{1}{2!} + \frac{1}{3!} + \dots + \frac{1}{n!} \right) \right] \leq \frac{1}{n+1}$$

$$0 \leq 1 - e^{-1} \left(1 + \frac{1}{1!} + \frac{1}{2!} + \dots + \frac{1}{n!} \right) \leq \frac{1}{(n+1)!}$$

(iv) As $n \rightarrow \infty$,
 $0 \leq \lim_{n \rightarrow \infty} \left[1 - e^{-1} \left(1 + \dots + \frac{1}{n!} \right) \right] \leq \lim_{n \rightarrow \infty} \frac{1}{(n+1)!}$

$$\text{But } \lim_{n \rightarrow \infty} \frac{1}{(n+1)!} = 0$$

$\therefore$

$$\therefore \lim_{n \rightarrow \infty} [1 - e^{-1} (1 + \frac{1}{1!} + \frac{1}{2!} + \dots + \frac{1}{n!})] = 0$$

$\therefore$ As $n \rightarrow \infty$,

$$(1 + \frac{1}{1!} + \frac{1}{2!} + \dots + \frac{1}{n!}) \rightarrow e$$

because $1 - e^{-1}(e) = 0.$